

Introduction to B-Tree & B+ Tree Data Structures

> **Definition:** A **B-Tree** is a self-balancing multi-way search tree of order m where each node can contain multiple keys and more than two children, designed specifically to optimize **disk read/write block accesses** in database indexing and file systems.

1. Detailed Technical Explanation

1. Properties of a B-Tree of Order m :

1. Every node has at most m children.
2. Every non-leaf node has at least $\lceil m/2 \rceil$ children.

2. B-Tree vs B+ Tree Comparison

B+ TREE:

Detailed Comparison Table:

Feature	B-Tree	B+ Tree
Leaf Nodes	Pointers to data blocks	Pointers to data blocks
Internal Nodes	Pointers to child nodes	Pointers to child nodes
Search Path	Search path leads to leaf node	Search path leads to leaf node
Range Query	Requires multiple disk accesses	Requires single disk access
Insertion	Requires multiple disk accesses	Requires single disk access
Deletion	Requires multiple disk accesses	Requires single disk access
Space Utilization	Higher space utilization	Lower space utilization
Performance	Higher performance	Lower performance

3. Core Concepts & Memory Keywords

- **Order m:** Maximum number of children a node can have.

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4. Must-Write Points for Exams

- B-Trees and B+ Trees reduce disk I/O by fitting large numbers of keys inside a single disk page block (e.g., 4KB or 8KB).

- **B+**

5. Quick Recall Flow

Multi-V